

ELECTRONICS LABORATORY

Experiment No:	1
Experiment Name:	Semiconductor Diodes
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1. OBJECTIVES

In this experiment, we're going to be learning about silicon and zener diodes. How they work at specific voltage levels, how they restrict the current, and the working principle of them when alternative current is applied.

2. EXPERIMENTAL RESULTS

In the first part of the experiment, we're going to use the diode in forward bias mode. We constructed a circuit that only consisted of a single silicon diode and started applying voltage, starting from 1 V all the way up to 10 V.

V_S	V_D	I_D
1 V	0.55 V	0.4 A
2 V	0.6 V	1.2 A
3 V	0.62 V	2 A
4 V	0.64 V	2.8 A
5 V	0.65 V	3.6 A
6 V	0.66 V	4.4 A
7 V	0.67 V	5.2 A
8 V	0.68 V	6 A
9 V	0.69 V	6.8 A
10 V	0.69 V	7.6 A

In the second part, we changed the terminals of the voltage source and started to increase the voltage and noting the results.

V_S	V_D	I_D
-1 V	-1.1 V	0 A
-2 V	-2.02 V	0 A
-3 V	-3.01 V	0 A
-4 V	-4.02 V	0 A
-5 V	-5.03 V	0 A
-6 V	-6 V	0 A
-7 V	-7.02V	0 A
-8 V	-8.01V	0 A
-9 V	-9.03 V	0 A
-10 V	-10 V	0 A

In the third part of the experiment, we construed the circuit as shown below and we tried to calculate the current flowing through the resistor. Unfortunately, we failed to measure the voltage as it was very small and the multimeter couldn't measure it. Instead, we used Ohm's Law to calculate the current running through the resistor.

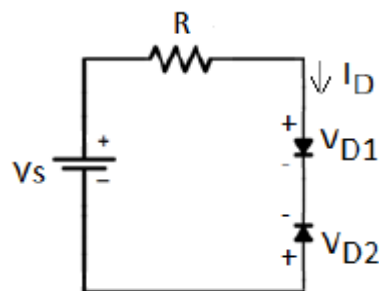


Figure 1

$$V = I * R$$

$$\frac{2*10^{-3}}{10^6} = I$$

$$2 * 10^{-9} A = I$$

Here, we found that the current running through the resistor is 2 nA which is very small for the multimeter to measure.

In the fourth and last part of the experiment, we connected an alternative voltage source to a zener diode in series to a resistor and we observed the voltage on the diode with the help of an oscilloscope.



Figure 2

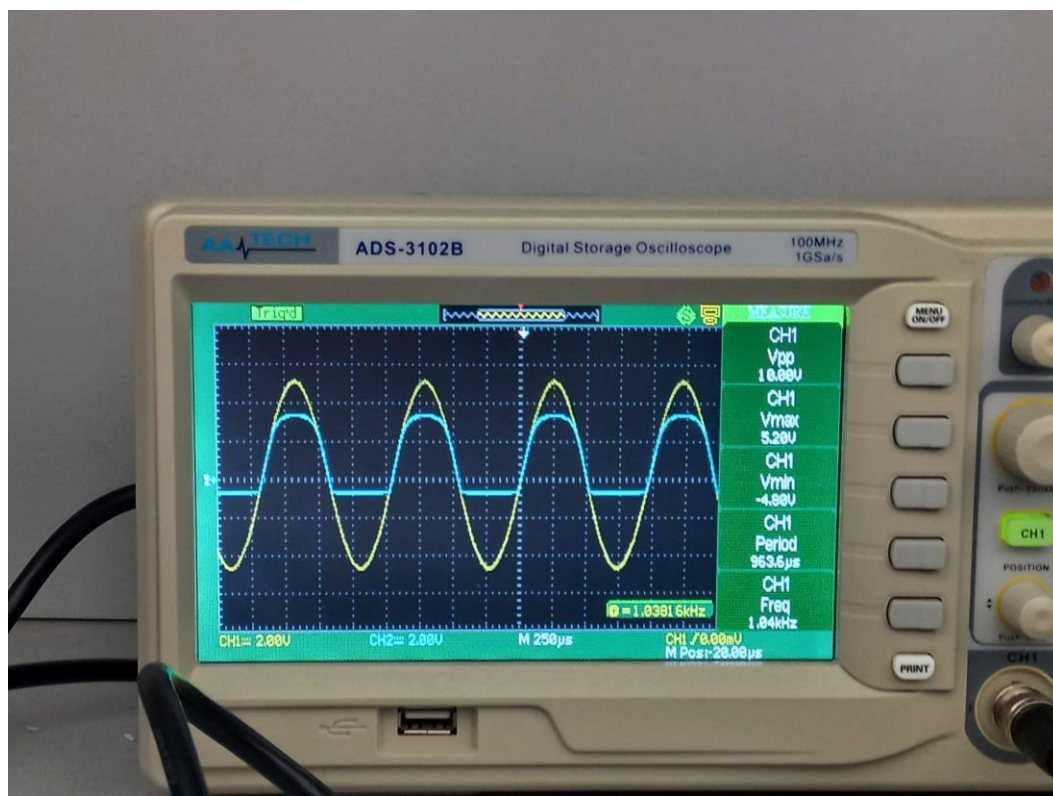


Figure 3

3. CONCLUSION & DISCUSSION

In the first part of the experiment, we see that the voltage on the diode is relatively small compared to the voltage on the resistor and because of this, current is able to pass through the diode thanks to the positive bias connection without too much loss.

In the second part, we observed that all of the voltage is applied to the diode and as a result of this no current passes through the circuit and then resistor. This occurred because of reverse bias connection.

In the third part, we connected two diodes opposite to each other. The multimeter couldn't measure the current running through the circuit. So, we calculated the current using Ohm's Law and found that the current is 2 nA which is very small and negligible. This happened because when we connected the diodes this way, it became always reverse biased which prevents the voltage source from conducting electricity to the circuit.

In the last part, we connected an alternative voltage source to a zener diode and observed the signals of the voltage on the diode on an oscilloscope. We found that the current sometimes passed through the diode and sometimes it didn't. The reason for this is, a zener diode passes current in both directions, but in different values. In Figure 3, we can see that it lets almost all of the current pass when negative voltage is applied, but when positive voltage is applied, it restricts most of the current passing through it, although some amount of it makes it through.

REFERENCES

- [1] IEEE Citation Style Guide, Available: <http://www.ijssst.info/info/IEEE-Citation-StyleGuide.pdf>, Access Date: October 2021